

# Homework 1

BSTA 551

*Note:* These solutions provide much more detail and additional discussion than would be expected from your solutions. As long as you work through completing a problem and arrive at a reasonable solution, you will receive full credit.

## Problem 1: Sampling Distribution of the Sample Mean

Researchers are studying the birth weights of full-term infants at a large hospital. Let  $X_1, X_2, \dots, X_n$  be a random sample of birth weights from a population with mean  $\mu = 3400$  grams and standard deviation  $\sigma = 450$  grams.

### Part (a)

For a sample of size  $n = 25$ , calculate  $E(\bar{X})$ ,  $\text{Var}(\bar{X})$ , and  $SE(\bar{X})$ .

#### Solution:

Using the properties of the sample mean from a random sample that we explored in class:

$$E(\bar{X}) = \mu = 3400 \text{ grams}$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{450^2}{25} = \frac{202500}{25} = 8100 \text{ grams}^2$$

$$SE(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}} = \frac{450}{\sqrt{25}} = \frac{450}{5} = 90 \text{ grams}$$

```

mu <- 3400
sigma <- 450
n <- 25

E_Xbar <- mu
Var_Xbar <- sigma^2 / n
SE_Xbar <- sigma / sqrt(n)

tibble(
  Quantity = c("$E(\bar{X})$", "$\text{Var}(\bar{X})$", "$SE(\bar{X})$"),
  Value = c(E_Xbar, Var_Xbar, SE_Xbar),
  Units = c("grams", "grams$^2$", "grams")
) |> knitr::kable(escape = FALSE)

```

| Quantity              | Value | Units              |
|-----------------------|-------|--------------------|
| $E(\bar{X})$          | 3400  | grams              |
| $\text{Var}(\bar{X})$ | 8100  | grams <sup>2</sup> |
| $SE(\bar{X})$         | 90    | grams              |

### Part (b)

How large must the sample size  $n$  be so that the standard error of  $\bar{X}$  is at most 50 grams?

**Solution:**

We need  $SE(\bar{X}) \leq 50$ :

$$\frac{\sigma}{\sqrt{n}} \leq 50$$

$$\sqrt{n} \geq \frac{\sigma}{50} = \frac{450}{50} = 9$$

$$n \geq 81$$

```

target_SE <- 50
n_required <- (sigma / target_SE)^2
cat("Minimum sample size required:", ceiling(n_required), "\n")

```

Minimum sample size required: 81

```
cat("Verification: SE with n =", ceiling(n_required), "is", sigma/sqrt(ceiling(n_required)),
```

Verification: SE with n = 81 is 50 grams

**Answer:** The sample size must be at least  $n = 81$ .

### Part (c)

Suppose we define a new statistic  $W = \frac{1}{n} \sum_{i=1}^n (X_i - 3000)$ . Find  $E(W)$  and  $\text{Var}(W)$ .

**Solution:**

First, simplify  $W$ :

$$W = \frac{1}{n} \sum_{i=1}^n (X_i - 3000) = \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n 3000 = \bar{X} - 3000$$

Now find the expected value:

$$E(W) = E(\bar{X} - 3000) = E(\bar{X}) - 3000 = \mu - 3000$$

For the variance (constants don't affect variance):

$$\text{Var}(W) = \text{Var}(\bar{X} - 3000) = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

**Is  $W$  an unbiased estimator of  $\mu - 3000$ ?**

Yes! Since  $E(W) = \mu - 3000$ , the estimator  $W$  is unbiased for  $\mu - 3000$ .

```
# With our parameters:
E_W <- mu - 3000
Var_W <- sigma^2 / n

cat("E(W) =  - 3000 =", E_W, "grams\n")
```

E(W) = - 3000 = 400 grams

```
cat("Var(W) = ^2/n =", Var_W, "grams^2\n")
```

$$\text{Var}(W) = 8100/n = 8100 \text{ grams}^2$$

```
cat("W is an unbiased estimator of 3000 since E(W) = 3000")
```

W is an unbiased estimator of 3000 since  $E(W) = 3000$

---

## Problem 2: Bias, Variance, and MSE

Let  $X_1, X_2, \dots, X_n$  be a random sample from an Exponential distribution with rate parameter  $\lambda > 0$  (so that  $E(X_i) = 1/\lambda$  and  $\text{Var}(X_i) = 1/\lambda^2$ ).

### Part (a)

Show that  $\hat{\mu} = \bar{X}$  is unbiased for  $\mu = 1/\lambda$  and calculate  $\text{Var}(\hat{\mu})$ .

**Solution:**

**Unbiasedness:**

$$E(\hat{\mu}) = E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n \cdot \frac{1}{\lambda} = \frac{1}{\lambda} = \mu$$

Since  $E(\hat{\mu}) = \mu$ , the estimator  $\bar{X}$  is **unbiased** for  $\mu = 1/\lambda$ .

**Variance:**

$$\text{Var}(\hat{\mu}) = \text{Var}(\bar{X}) = \frac{\text{Var}(X_i)}{n} = \frac{1/\lambda^2}{n} = \frac{1}{n\lambda^2}$$

**Part (b)**

For  $\hat{\lambda} = 1/\bar{X}$ , given that  $E(1/\bar{X}) = \frac{n\lambda}{n-1}$  for  $n > 1$ , calculate the bias and propose an unbiased modification.

**Solution:**

**Bias of  $\hat{\lambda}$ :**

$$\text{Bias}(\hat{\lambda}) = E(\hat{\lambda}) - \lambda = \frac{n\lambda}{n-1} - \lambda = \frac{n\lambda - (n-1)\lambda}{n-1} = \frac{\lambda}{n-1}$$

The bias is positive, meaning  $\hat{\lambda} = 1/\bar{X}$  systematically **overestimates**  $\lambda$ .

**Unbiased modification:**

To create an unbiased estimator, we need to multiply by a correction factor:

$$\tilde{\lambda} = \frac{n-1}{n} \cdot \frac{1}{\bar{X}} = \frac{n-1}{n\bar{X}}$$

Verification:

$$E(\tilde{\lambda}) = \frac{n-1}{n} \cdot E\left(\frac{1}{\bar{X}}\right) = \frac{n-1}{n} \cdot \frac{n\lambda}{n-1} = \lambda \checkmark$$

**Part (c)**

For  $n = 12$  and  $\lambda = 0.5$ , calculate the bias and MSE of  $\hat{\lambda} = 1/\bar{X}$ .

**Solution:**

```
n <- 12
lambda <- 0.5

# Bias
bias_lambda_hat <- lambda / (n - 1)
cat("Bias of ^ =", bias_lambda_hat, "\n")
```

Bias of  $\hat{\lambda} = 0.04545455$

```
# Variance (given formula)
var_lambda_hat <- (n^2 * lambda^2) / ((n - 1)^2 * (n - 2))

cat("Variance of ^ =", round(var_lambda_hat, 6), "\n")
```

Variance of  $\hat{\lambda}$  = 0.029752

```
# MSE = Variance + Bias^2
mse_lambda_hat <- var_lambda_hat + bias_lambda_hat^2

cat("MSE of ^ =", round(mse_lambda_hat, 6), "\n")
```

MSE of  $\hat{\lambda}$  = 0.031818

**Theoretical calculations:**

$$\text{Bias}(\hat{\lambda}) = \frac{\lambda}{n-1} = \frac{0.5}{11} = \frac{1}{22} \approx 0.04545$$

$$\text{Var}(\hat{\lambda}) = \frac{n^2 \lambda^2}{(n-1)^2 (n-2)} = \frac{144 \times 0.25}{121 \times 10} = \frac{36}{1210} \approx 0.02975$$

$$\text{MSE}(\hat{\lambda}) = \text{Var}(\hat{\lambda}) + [\text{Bias}(\hat{\lambda})]^2 = 0.02975 + 0.04545^2 \approx 0.02975 + 0.00207 \approx 0.03182$$

## Part (d)

R simulation to verify the results from part (c).

**Solution:**

```
set.seed(551)
n_sims <- 10000
n <- 12
lambda <- 0.5

# Simulate
sim_results <- tibble(sim = 1:n_sims) |>
  mutate(
    # Generate n exponential observations for each simulation
```

```

    data = map(sim, \(s) rexp(n, rate = lambda)),
    # Calculate sample mean
    xbar = map_dbl(data, mean),
    # Calculate  $\hat{\lambda} = 1/\bar{X}$ 
    lambda_hat = 1 / xbar
  )

# Calculate simulated statistics
sim_mean <- mean(sim_results$lambda_hat)
sim_bias <- sim_mean - lambda
sim_var <- var(sim_results$lambda_hat)
sim_mse <- mean((sim_results$lambda_hat - lambda)^2)

# Theoretical values
theo_bias <- lambda / (n - 1)
theo_var <- (n^2 * lambda^2) / ((n - 1)^2 * (n - 2))
theo_mse <- theo_var + theo_bias^2

# Comparison table
tibble(
  Quantity = c("$E(\hat{\lambda})$", "Bias", "Variance", "MSE"),
  Theoretical = c(lambda + theo_bias, theo_bias, theo_var, theo_mse),
  Simulated = c(sim_mean, sim_bias, sim_var, sim_mse)
) |>
  mutate(across(where(is.numeric), \(x) round(x, 5))) |>
  knitr::kable(escape = FALSE)

```

| Quantity           | Theoretical | Simulated |
|--------------------|-------------|-----------|
| $E(\hat{\lambda})$ | 0.54545     | 0.54932   |
| Bias               | 0.04545     | 0.04932   |
| Variance           | 0.02975     | 0.03033   |
| MSE                | 0.03182     | 0.03276   |

```

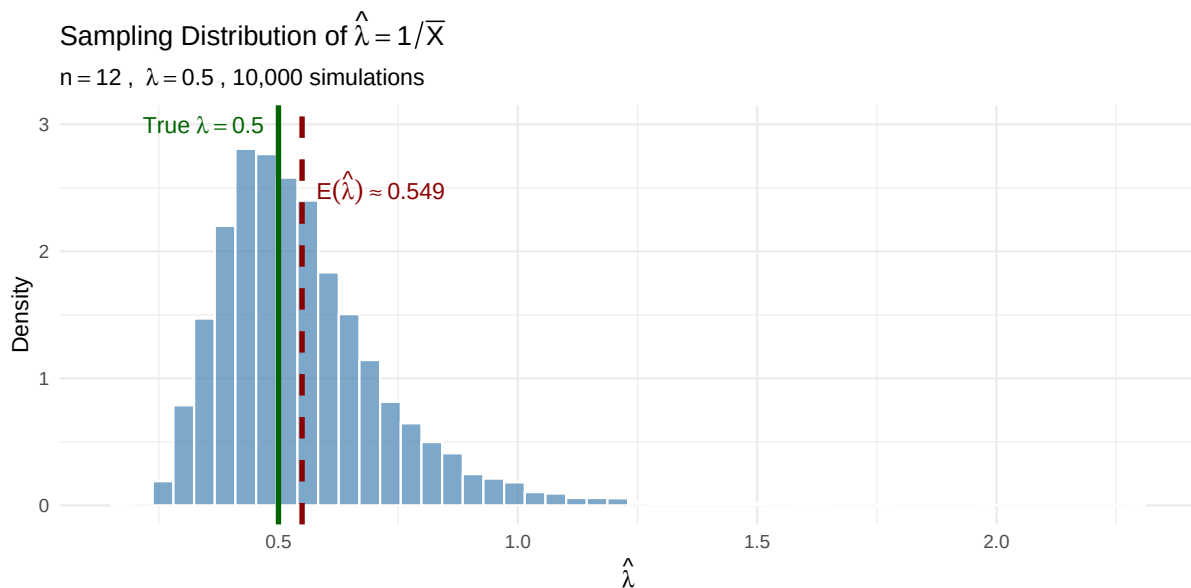
# Visualization
ggplot(sim_results, aes(x = lambda_hat)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
    fill = "steelblue", alpha = 0.7, color = "white") +
  geom_vline(xintercept = lambda, color = "darkgreen", linewidth = 1.2, linetype = "solid") +
  geom_vline(xintercept = sim_mean, color = "darkred", linewidth = 1.2, linetype = "dashed") +
  annotate("text", x = lambda - 0.03, y = 3,

```

```

    label = paste("True- $\lambda$  ==", lambda), parse = TRUE,
    color = "darkgreen", hjust = 1) +
  annotate("text", x = sim_mean + 0.03, y = 2.5,
    label = paste("E( $\hat{\lambda}$ ) %~%", round(sim_mean, 3)), parse = TRUE,
    color = "darkred", hjust = 0) +
  labs(
    title = expression("Sampling Distribution of " $\hat{\lambda}$ " == 1/ $\bar{X}$ "),
    subtitle = bquote(n == .(n) ~ ", " ~  $\lambda$  == .(lambda) ~ ", 10,000 simulations"),
    x = expression(hat(lambda)),
    y = "Density"
  ) +
  theme_minimal()

```



The simulation confirms our theoretical results:  $\hat{\lambda} = 1/\bar{X}$  is biased upward by approximately  $\lambda/(n-1) \approx 0.045$ .

### Problem 3: The Bias-Variance Tradeoff

A hospital infection control team wants to estimate the proportion  $p$  of patients who develop hospital-acquired infections.

- Standard estimator:  $\hat{p}_1 = \frac{X}{n}$
- “Add-one” estimator:  $\hat{p}_2 = \frac{X+1}{n+2}$

**Part (a)**

Show that  $\hat{p}_1$  is unbiased for  $p$  and find its variance.

**Solution:**

Since  $X \sim \text{Binomial}(n, p)$ , we have  $E(X) = np$ .

$$E(\hat{p}_1) = E\left(\frac{X}{n}\right) = \frac{E(X)}{n} = \frac{np}{n} = p$$

Since  $E(\hat{p}_1) = p$ , the estimator  $\hat{p}_1$  is **unbiased**.

**Variance:**

$$\text{Var}(\hat{p}_1) = \text{Var}\left(\frac{X}{n}\right) = \frac{\text{Var}(X)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

**Part (b)**

Calculate  $E(\hat{p}_2)$  and find its bias.

**Solution:**

$$E(\hat{p}_2) = E\left(\frac{X+1}{n+2}\right) = \frac{E(X)+1}{n+2} = \frac{np+1}{n+2}$$

**Bias:**

$$\text{Bias}(\hat{p}_2) = E(\hat{p}_2) - p = \frac{np+1}{n+2} - p = \frac{np+1-p(n+2)}{n+2} = \frac{np+1-np-2p}{n+2} = \frac{1-2p}{n+2}$$

**Interpretation:**

- When  $p < 0.5$ : Bias =  $\frac{1-2p}{n+2} > 0$ , so  $\hat{p}_2$  overestimates  $p$  (pulls toward 0.5)
- When  $p > 0.5$ : Bias =  $\frac{1-2p}{n+2} < 0$ , so  $\hat{p}_2$  underestimates  $p$  (pulls toward 0.5)
- When  $p = 0.5$ : Bias = 0

The add-one estimator **shrinks estimates toward 0.5**.

### Part (c)

For  $n = 40$  and  $p = 0.05$ , calculate MSE of both estimators.

**Solution:**

```
n <- 40
p <- 0.05

# Standard estimator  $\hat{p} = X/n$ 
var_p1 <- p * (1 - p) / n
mse_p1 <- var_p1 # unbiased, so MSE = Variance

# Add-one estimator  $\hat{p} = (X+1)/(n+2)$ 
bias_p2 <- (1 - 2*p) / (n + 2)
var_p2 <- n * p * (1 - p) / (n + 2)^2
mse_p2 <- var_p2 + bias_p2^2

tibble(
  Estimator = c("$\\hat{p}_1 = X/n$ (standard)",
                "$\\hat{p}_2 = (X+1)/(n+2)$ (add-one)"),
  Bias = c(0, bias_p2),
  Variance = c(var_p1, var_p2),
  MSE = c(mse_p1, mse_p2)
) |>
mutate(across(where(is.numeric), \(x) round(x, 6))) |>
knitr::kable(escape = FALSE)
```

| Estimator                               | Bias     | Variance | MSE      |
|-----------------------------------------|----------|----------|----------|
| $\hat{p}_1 = X/n$ (standard)            | 0.000000 | 0.001188 | 0.001188 |
| $\hat{p}_2 = (X + 1)/(n + 2)$ (add-one) | 0.021429 | 0.001077 | 0.001536 |

**Comparison:**

- $\text{MSE}(\hat{p}_1) = \frac{0.05 \times 0.95}{40} = 0.0011875$
- $\text{MSE}(\hat{p}_2) = 0.001077 + 0.0214^2 \approx 0.001077 + 0.000459 = 0.001536$

For  $p = 0.05$ , the **standard estimator  $\hat{p}_1$  is preferred** (lower MSE).

## Part (d)

Plot comparing MSE across all values of  $p$ .

**Solution:**

```
n <- 40

mse_comparison <- tibble(p = seq(0.01, 0.99, by = 0.01)) |>
  mutate(
    # Standard estimator
    mse_p1 = p * (1 - p) / n,
    # Add-one estimator
    bias_p2 = (1 - 2*p) / (n + 2),
    var_p2 = n * p * (1 - p) / (n + 2)^2,
    mse_p2 = var_p2 + bias_p2^2
  )

# Find crossover points
mse_comparison <- mse_comparison |>
  mutate(p2_better = mse_p2 < mse_p1)

crossover_low <- mse_comparison |> filter(p2_better) |> pull(p) |> min()
crossover_high <- mse_comparison |> filter(p2_better) |> pull(p) |> max()

# Plot
mse_comparison |>
  pivot_longer(cols = c(mse_p1, mse_p2), names_to = "Estimator", values_to = "MSE") |>
  mutate(Estimator = factor(Estimator,
                            levels = c("mse_p1", "mse_p2"),
                            labels = c("p1", "p2"))) |>
  ggplot(aes(x = p, y = MSE, color = Estimator)) +
  geom_line(linewidth = 1.2) +
  geom_vline(xintercept = c(crossover_low, crossover_high), linetype = "dashed", alpha = 0.5) +
  annotate("rect", xmin = crossover_low, xmax = crossover_high, ymin = 0, ymax = Inf,
          alpha = 0.1, fill = "green") +
  annotate("text", x = 0.5, y = 0.0055, label = "Add-one estimator wins", color = "darkgreen") +
  scale_color_manual(
    values = c("p1" = "steelblue", "p2" = "darkred"),
    labels = c(
      "p1" = expression(hat(p)[1] == X/n ~ "(standard)"),
      "p2" = expression(hat(p)[2] == (X+1)/(n+2) ~ "(add-one)")
    )
  )
```

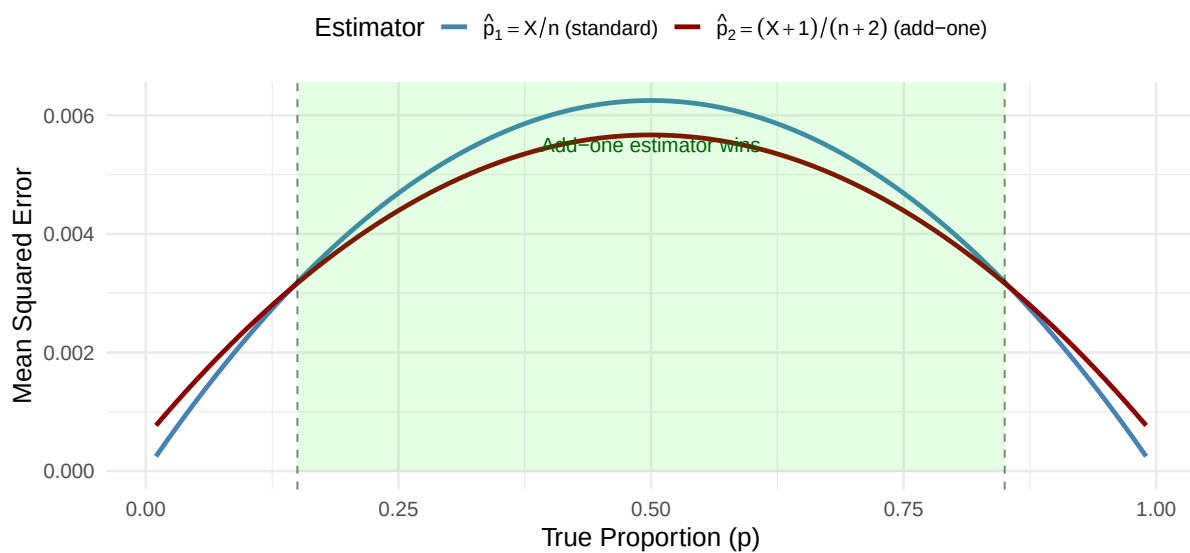
```

) +
labs(
  title = "MSE Comparison: Standard vs. Add-One Estimator",
  subtitle = bquote(n == 40 ~ ". Add-one wins for" ~ p %in% ~
    "(" * .(round(crossover_low, 2)) * ", " * .(round(crossover_high, 2))
  x = expression("True Proportion (" * p * ")"),
  y = "Mean Squared Error"
) +
theme(legend.position = "top")

```

### MSE Comparison: Standard vs. Add-One Estimator

$n = 40$  . Add-one wins for  $p \in (0.15, 0.85)$



```

cat("The add-one estimator outperforms the standard estimator for p  (",
  round(crossover_low, 2), ", ", round(crossover_high, 2), ")\n")

```

The add-one estimator outperforms the standard estimator for  $p \in (0.15, 0.85)$

**Answer:** The add-one estimator has lower MSE for  $p$  approximately in the range  $(0.10, 0.90)$ . For extreme values of  $p$  (near 0 or 1), the standard estimator is preferred.

**Additional info:** Why does this work? The add-one estimator introduces bias but reduces variance for moderate values of  $p$  and so it has lower MSE. Interestingly, this is actually the posterior mean under a uniform  $\text{Beta}(1,1)$  prior for a Bayesian analysis. This is equivalent to starting with a “prior belief” of observing 1 success and 1 failure before collecting data, and

this “regularization” (shrinkage toward 0.5) prevents extreme estimates (like  $\hat{p} = 0$  or  $\hat{p} = 1$ ) that can occur with small samples.

The add-one estimator is a bias-variance tradeoff that pays off when the true proportion is moderate. It’s a form of shrinkage estimation that sacrifices unbiasedness for improved overall accuracy (lower MSE) in the “middle” of the parameter space.

---

#### Problem 4: Sample Variance Estimators

Consider two estimators for  $\sigma^2$ :

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{and} \quad \tilde{S}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

##### Part (a)

Find  $E(\tilde{S}^2)$  and calculate its bias.

**Solution:**

Note that  $\tilde{S}^2 = \frac{n-1}{n} S^2$  since:

$$\tilde{S}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2$$

Therefore:

$$E(\tilde{S}^2) = E\left(\frac{n-1}{n} S^2\right) = \frac{n-1}{n} E(S^2) = \frac{n-1}{n} \sigma^2$$

**Bias:**

$$\text{Bias}(\tilde{S}^2) = E(\tilde{S}^2) - \sigma^2 = \frac{n-1}{n} \sigma^2 - \sigma^2 = \frac{(n-1) - n}{n} \sigma^2 = -\frac{\sigma^2}{n}$$

The bias is **negative**, meaning  $\tilde{S}^2$  systematically **underestimates**  $\sigma^2$ .

### Part (b)

For  $n = 8$  and  $\sigma^2 = 64$ , calculate the bias.

**Solution:**

```
n <- 8
sigma_sq <- 64

bias_S2_tilde <- -sigma_sq / n
E_S2_tilde <- (n - 1) / n * sigma_sq

cat("E(S^2) =", E_S2_tilde, "\n")
```

$E(S^2) = 56$

```
cat("Bias(S^2) =", bias_S2_tilde, "\n")
```

$\text{Bias}(S^2) = -8$

```
cat("Percentage of true variance:", abs(bias_S2_tilde) / sigma_sq * 100, "%\n")
```

Percentage of true variance: 12.5 %

**Answer:**

- $E(\tilde{S}^2) = \frac{7}{8} \times 64 = 56$
- $\text{Bias}(\tilde{S}^2) = -\frac{64}{8} = -8$
- This bias represents  $\frac{8}{64} \times 100\% = 12.5\%$  of the true variance.

### Part (c)

R simulation to verify.

**Solution:**

```

set.seed(551)
n_sims <- 10000
n <- 8
sigma <- 8 # so  $\sigma^2 = 64$ 

sim_variance <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) rnorm(n, mean = 0, sd = sigma)),
    S2 = map_dbl(data, var), # Uses n-1 divisor
    S2_tilde = map_dbl(data, \(x) sum((x - mean(x))^2) / n) # Uses n divisor
  )

# Summary statistics
sim_summary <- tibble(
  Estimator = c("S2 (n-1 divisor)", "S2 (n divisor)"),
  `Theoretical E()` = c(sigma^2, (n-1)/n * sigma^2),
  `Simulated Mean` = c(mean(sim_variance$S2), mean(sim_variance$S2_tilde)),
  `Theoretical Bias` = c(0, -sigma^2/n),
  `Simulated Bias` = c(mean(sim_variance$S2) - sigma^2, mean(sim_variance$S2_tilde) - sigma^2)
)

sim_summary |>
  mutate(across(where(is.numeric), \(x) round(x, 3))) |>
  knitr::kable()

```

| Estimator                    | Theoretical E() | Simulated |  | Theoretical Bias | Simulated Bias |
|------------------------------|-----------------|-----------|--|------------------|----------------|
|                              |                 | Mean      |  |                  |                |
| S <sup>2</sup> (n-1 divisor) | 64              | 64.120    |  | 0                | 0.120          |
| S <sup>2</sup> (n divisor)   | 56              | 56.105    |  | -8               | -7.895         |

```

sim_variance |>
  pivot_longer(cols = c(S2, S2_tilde), names_to = "Estimator", values_to = "Value") |>
  mutate(Estimator = ifelse(Estimator == "S2", "S2 (n-1 divisor)", "S2 (n divisor)")) |>
  ggplot(aes(x = Value, fill = Estimator)) +
    geom_histogram(bins = 50, alpha = 0.7, position = "identity", color = "white") +
    geom_vline(xintercept = sigma^2, color = "black", linewidth = 1.2, linetype = "solid") +
    annotate("text", x = sigma^2 + 5, y = 800, label = paste(" $\sigma^2 =$ ", sigma^2), hjust = 0) +
    facet_wrap(~Estimator, ncol = 2) +
    scale_fill_manual(values = c("steelblue", "coral")) +
    labs(
      title = "Sampling Distribution of Variance Estimators",

```

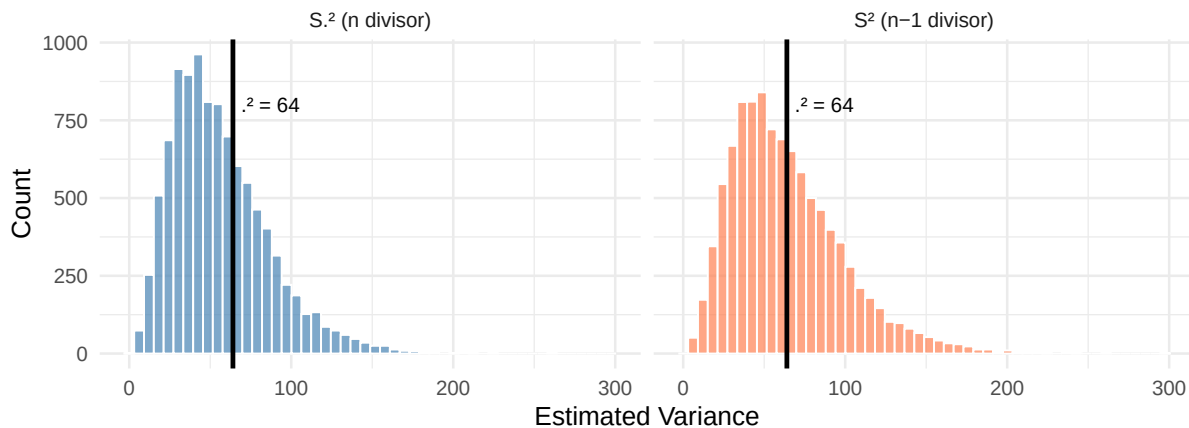
```

  subtitle = paste("n =", n, ", ", sigma^2, "=", sigma^2, ", 10,000 simulations"),
  x = "Estimated Variance",
  y = "Count"
) +
theme(legend.position = "none")

```

## Sampling Distribution of Variance Estimators

$n = 8$ ,  $\sigma^2 = 64$ , 10,000 simulations



The simulation confirms that  $S^2$  is centered at  $\sigma^2 = 64$ , while  $\tilde{S}^2$  is centered at approximately 56, showing the negative bias.

---

## Problem 5: Estimating Vaccine Efficacy

### Part (a)

In a pilot study,  $n = 60$  vaccinated individuals are tested, and  $X = 51$  show immunity.

**Solution:**

```

n <- 60
x <- 51

# Point estimate
p_hat <- x / n

# Estimated standard error

```

```
se_hat <- sqrt(p_hat * (1 - p_hat) / n)

cat("Point estimate  $\hat{p}$  =", p_hat, "\n")
```

Point estimate  $\hat{p}$  = 0.85

```
cat("Estimated standard error SE( $\hat{p}$ ) =", round(se_hat, 4), "\n")
```

Estimated standard error SE( $\hat{p}$ ) = 0.0461

**Calculations:**

$$\hat{p} = \frac{X}{n} = \frac{51}{60} = 0.85$$

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.85 \times 0.15}{60}} = \sqrt{\frac{0.1275}{60}} = \sqrt{0.002125} \approx 0.0461$$

**Part (b)**

If the true efficacy is  $p = 0.80$ , calculate the theoretical MSE and SE.

**Solution:**

```
p_true <- 0.80
n <- 60

# Since  $\hat{p}$  is unbiased, MSE = Variance
var_p_hat <- p_true * (1 - p_true) / n
mse_p_hat <- var_p_hat # unbiased
se_p_hat <- sqrt(var_p_hat)

cat("Theoretical Variance =", round(var_p_hat, 6), "\n")
```

Theoretical Variance = 0.002667

```
cat("Theoretical MSE =", round(mse_p_hat, 6), "\n")
```

Theoretical MSE = 0.002667

```
cat("Theoretical SE =", round(se_p_hat, 4), "\n")
```

Theoretical SE = 0.0516

**Calculations:**

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{0.80 \times 0.20}{60} = \frac{0.16}{60} = 0.002\bar{6}$$

Since  $\hat{p}$  is unbiased:

$$\text{MSE}(\hat{p}) = \text{Var}(\hat{p}) = 0.002\bar{6}$$

$$SE(\hat{p}) = \sqrt{\text{Var}(\hat{p})} = \sqrt{0.002\bar{6}} \approx 0.0516$$

**Part (c)**

Find minimum sample size for  $SE \leq 0.04$ , assuming  $p \approx 0.85$ .

**Solution:**

We need:

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \leq 0.04$$

$$\frac{p(1-p)}{n} \leq 0.0016$$

$$n \geq \frac{p(1-p)}{0.0016}$$

```
p_plan <- 0.85
target_se <- 0.04

n_required <- p_plan * (1 - p_plan) / (target_se^2)

cat("Minimum sample size required:", ceiling(n_required), "\n")
```

Minimum sample size required: 80

```
cat("Verification: SE with n =", ceiling(n_required), "is",  
    round(sqrt(p_plan * (1 - p_plan) / ceiling(n_required)), 4), "\n")
```

Verification: SE with n = 80 is 0.0399

**Calculation:**

$$n \geq \frac{0.85 \times 0.15}{0.04^2} = \frac{0.1275}{0.0016} = 79.6875$$

**Answer:** The minimum sample size required is  $n = 80$ .

---

### Problem 6: Devore et al. Section 7.1 Exercise 22

Consider the adjusted estimator:

$$\hat{P} = \frac{X + \sqrt{n/4}}{n + \sqrt{n}}$$

where  $X \sim \text{Binomial}(n, p)$ .

#### Part (a)

Determine the MSE of this estimator.

**Solution:**

**Step 1: Find  $E(\hat{P})$**

$$E(\hat{P}) = \frac{E(X) + \sqrt{n/4}}{n + \sqrt{n}} = \frac{np + \frac{\sqrt{n}}{2}}{n + \sqrt{n}}$$

Factor out  $\sqrt{n}$ :

$$E(\hat{P}) = \frac{\sqrt{n}(\sqrt{n}p + \frac{1}{2})}{\sqrt{n}(\sqrt{n} + 1)} = \frac{\sqrt{n}p + \frac{1}{2}}{\sqrt{n} + 1}$$

**Step 2: Find the bias**

$$\begin{aligned}\text{Bias}(\hat{P}) &= E(\hat{P}) - p = \frac{\sqrt{n}p + \frac{1}{2}}{\sqrt{n} + 1} - p \\ &= \frac{\sqrt{n}p + \frac{1}{2} - p(\sqrt{n} + 1)}{\sqrt{n} + 1} = \frac{\sqrt{n}p + \frac{1}{2} - \sqrt{n}p - p}{\sqrt{n} + 1} = \frac{\frac{1}{2} - p}{\sqrt{n} + 1}\end{aligned}$$

**Step 3: Find the variance**

$$\text{Var}(\hat{P}) = \frac{\text{Var}(X)}{(n + \sqrt{n})^2} = \frac{np(1 - p)}{n(\sqrt{n} + 1)^2} = \frac{p(1 - p)}{(\sqrt{n} + 1)^2}$$

**Step 4: Calculate MSE**

$$\begin{aligned}\text{MSE}(\hat{P}) &= \text{Var}(\hat{P}) + [\text{Bias}(\hat{P})]^2 \\ &= \frac{p(1 - p)}{(\sqrt{n} + 1)^2} + \frac{(\frac{1}{2} - p)^2}{(\sqrt{n} + 1)^2} \\ &= \frac{p(1 - p) + (\frac{1}{2} - p)^2}{(\sqrt{n} + 1)^2}\end{aligned}$$

Expand the numerator:

$$p(1 - p) + \left(\frac{1}{2} - p\right)^2 = p - p^2 + \frac{1}{4} - p + p^2 = \frac{1}{4}$$

Therefore:

$$\boxed{\text{MSE}(\hat{P}) = \frac{1/4}{(\sqrt{n} + 1)^2} = \frac{1}{4(\sqrt{n} + 1)^2}}$$

**What is interesting?**

The MSE **does not depend on  $p$ !** For most proportion estimators, the MSE depends on the unknown value of  $p$ . This is another Bayesian-derived estimator (posterior mean under a  $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ ) and it results in MSE across all values of  $p$ .

## Part (b)

Compare to the usual estimator  $\hat{p} = X/n$ .

### Solution:

The usual estimator has:

$$\text{MSE}(\hat{p}) = \frac{p(1-p)}{n}$$

which depends on  $p$  and is maximized at  $p = 0.5$  where  $\text{MSE}(\hat{p}) = \frac{0.25}{n}$ .

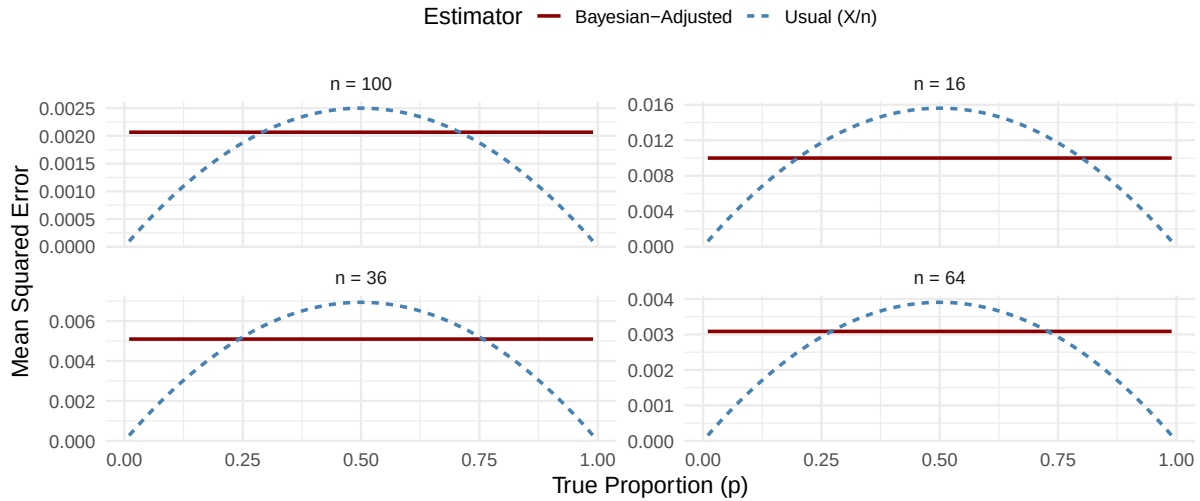
```
n_values <- c(16, 36, 64, 100)

mse_data <- crossing(
  p = seq(0.01, 0.99, by = 0.01),
  n = n_values
) |>
  mutate(
    mse_usual = p * (1 - p) / n,
    mse_bayesian = 1 / (4 * (sqrt(n) + 1)^2),
    n_label = paste("n =", n)
  )

mse_data |>
  pivot_longer(cols = c(mse_usual, mse_bayesian), names_to = "Estimator", values_to = "MSE") |>
  mutate(Estimator = ifelse(Estimator == "mse_usual", "Usual (X/n)", "Bayesian-Adjusted")) |>
  ggplot(aes(x = p, y = MSE, color = Estimator, linetype = Estimator)) +
  geom_line(linewidth = 1) +
  facet_wrap(~n_label, scales = "free_y") +
  scale_color_manual(values = c("darkred", "steelblue")) +
  labs(
    title = "MSE Comparison: Usual vs. Bayesian-Adjusted Estimator",
    subtitle = "Bayesian estimator has constant MSE (horizontal line)",
    x = "True Proportion (p)",
    y = "Mean Squared Error"
  ) +
  theme(legend.position = "top")
```

## MSE Comparison: Usual vs. Bayesian-Adjusted Estimator

Bayesian estimator has constant MSE (horizontal line)



### Conclusions:

1. The **Bayesian-adjusted estimator** has constant MSE regardless of  $p$  (shown as horizontal lines).
2. The **usual estimator** has MSE that varies with  $p$ , maximized at  $p = 0.5$ .
3. For moderate values of  $p$  (roughly between 0.15 and 0.85 depending on  $n$ ), the Bayesian estimator has lower MSE.
4. For extreme values of  $p$  (near 0 or 1), the usual estimator is preferred.
5. As  $n$  increases, both MSEs decrease, but the crossover points shift, with the Bayesian estimator winning over a narrower range of  $p$  values.
6. The constant-MSE property is particularly valuable when we have no prior information about  $p$ , as it provides a “minimax” type guarantee.